The Failure Point of Systems of Linearized Differential Equations

How far from equilibrium does the linearization of a two-variable energy-balance climate model remain a valid predictor of the system's behaviour?

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1 Introduction

2 Modelling

With the aim of understanding how the number of rounds affect the orderedness of the cards in play, I will be modelling the following relationship:

x-axis: the number of rounds played (independent variable)
y-axis: the cards flipped from the draw pile (dependent variable)

I am using the variable of cards flipped from the draw pile because this is a key indicator that isolates the orderedness of only the cards in play rather than assessing the orderedness of the whole deck, reducing the scope and therefore increasing the validity. The values found are included in tabulated form in the Appendix, and in graph form below: